

Roll No. _____

Name of the Student _____

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION NOV-2023

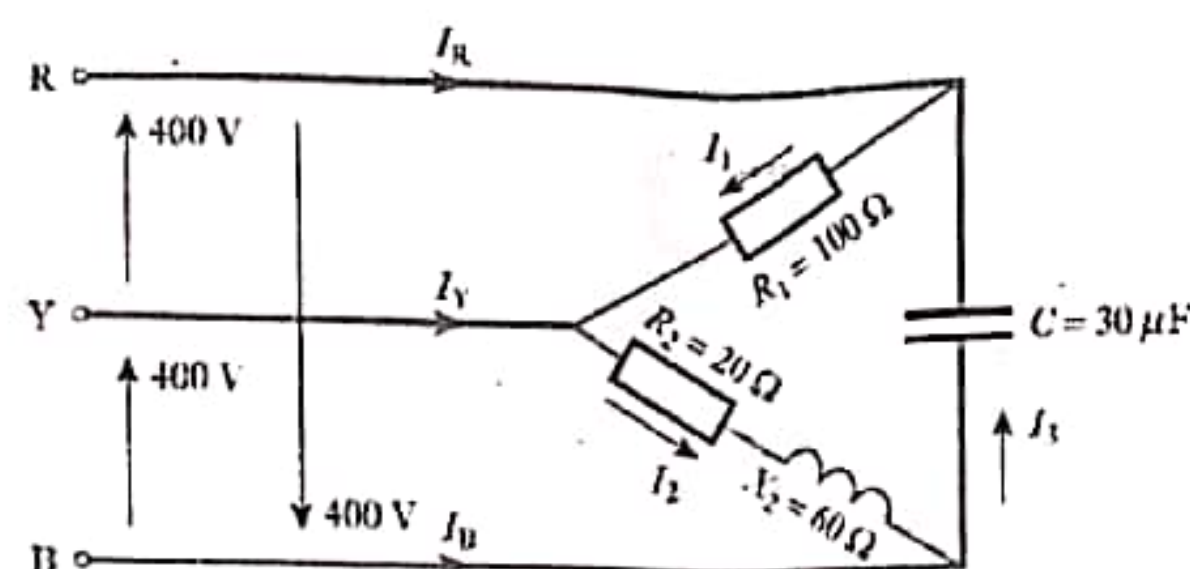
Branch: **Electrical Engg.**Subject Code: **EE-212**Max Marks: **50**Course: **B.Tech**Semester **III**Subject Name : **Electrical Machines-I**Time : **180 minutes**NOTE: **Answer all Questions****Marks**

1. a. Prove that the power in a balanced 3-phase circuit can be deduced from the readings of two watt-meters. Draw the relevant connection diagram and phasor diagram.

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b. A delta-connected load is arranged as shown in Fig. below. The supply voltage is 400 V at 50 Hz. Calculate:
 (a) the phase currents (b) the line currents. Draw the phasor diagram.

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2. a. When the three identical star connected coils are supplied with 440 V, 50 Hz, 3-Phase supply, the 1-phase wattmeter whose current coil is connected in line R and pressure coil across the phase R and neutral reads 6 kW and the ammeter connected in R-phase reads 30 A. Assuming RYB phase sequence find (i) Resistance and reactance of the coil (ii) Power factor of the load (iii) Reactive power of load

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b. Draw and explain the Sumpner's test on single phase transformer

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3. a. A three phase step down transformer is connected to 6600 V on the primary side. The ratio of turns per phase is 10 and the line current from the mains is 25 A. Find secondary line voltage, line current and output if the transformer is (a) star-star (b) delta-delta (iii) star-delta (iv) delta-delta. Neglect losses.

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b. Draw the physical connection and phasor diagrams of the following transformer connections: (i) Yd11 (ii) Dy1 (iii) Dd6

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4. a. A 230 V DC shunt motor has armature resistance of 0.4 ohm and field resistance of 115 ohms. The motor drives a constant torque and takes an armature current of 20 A and 800 rpm. If motor speed is to be raised from 800 rpm to 1000 rpm, find the resistance that must be inserted in the shunt field circuit. Assume magnetizing characteristic curve to be linear.

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b. Explain the effect of armature reaction in a DC shunt generator. How are its demagnetizing and cross-magnetizing ampere turns calculated?

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5. a. Draw and explain the internal and external characteristics curves of a dc series motor. Also write the relevant equations.

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b. A D.C. shunt motor with armature circuit resistance of 0.1 ohms runs at 1000 rpm while taking an armature current of 100A from 230 V dc source. The friction and windage losses are 300 Watt, no load core losses are 1200 Watt, total I²R losses is 2500 Watts and stray losses equals 1% of output. Find the shaft torque of motor and its efficiency.

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